Section 2: Data Overview

Cy Coldiron

3.1 Tweet Data Overview

The dataset used for classifying tweets is Tweets of Congress (maintained by Alex Litel), which contains all tweets (original, retweets, and quoted retweets) from the official accounts of U.S. congressional members between June 2017 and January 2023. The data was generated via an open-source, serverless application that interfaces with the Twitter API, querying for new tweets at the top of each hour. Each observation (tweet) is stored as an object (with one JSON file corresponding to each day). This automated pipeline ensures comprehensive, continuous coverage of congressional Twitter activity during that period, with accounts updated through routine synchronization with legislator metadata. For instance, this system automatically adds newly sworn-in members to the metadata and removes non-members (i.e., those who lost an election, resigned, etc.)—thereby reflecting Congress's true composition at each point in time.

Although more recent data (2024-2025) would provide greater relevance and utility, especially with the passage of the Big Beautiful Bill (BBB), the cost of x.com's current API access—which changed following Elon Musk's acquisition—is beyond this project's financial resources. For reference, under current pricing structures, replicating a dataset of a similar size (roughly six million tweets) would cost between \$30,000 and \$42,000.

While the raw dataset itself includes party committee accounts (oversight, armed services, education, etc.) and institutional accounts (housedems, senateGOP), our project focuses exclusively on the rhetorical patterns of congressional members. Thus, we remove all accounts not directly tied to an individual congressional member, finalizing the dataset at roughly 3.6 million tweets.

3.2 Tweet classification Strategy

To identify deficit-related tweets, I implemented a two-tier keyword strategy in Python (pandas). The full code can be seen at the projects GitHub Repo.

Tier 1 – Anchor Keywords. Tweets were flagged as deficit-related if they contained unambiguous anchor phrases (e.g., *"national debt," "balanced budget," "fiscal responsibility"*). These strong keywords identify explicit deficit references with high confidence.

Tier 2 – Contextual Expansion. Tweets containing weaker terms (e.g., *"debt"*) were only flagged if they co-occurred with contextual phrases (e.g., *"federal," "budget," "tax," "spending"*).

The logic of my labeling framework is to balance the two key countervailing forces in text classification: precision and coverage. While a larger keyword dictionary would capture more subtle deficit references, it would also reduce precision by inflating the number of false positives. For instance, including phrases like 'trillions of dollars' may capture references that would otherwise be missed, it also risks falsely tagging a growing number of unrelated tweets, such as discussions of income inequality or GDP growth. By contrast, a smaller dictionary would reduce false positives but would also diminish overall coverage, leading to more deficit related tweets going un-labeled.

Illustrative Example:

Figures 1 and 2 demonstrates the labeling logic. Rep. Alexandria Ocasio-Cortez's reference to "student loan debt" is not flagged: "debt" is a weak keyword but lacks an accompanying context term. By contrast, Speaker Kevin McCarthy's reference to the "national debt" matches a strong anchor keyword and is thus labeled as deficit related.

(Note: Speaker McCarthy's tweet was published outside our anlaytical timeframe (January 2023) but still serves as an informative example).

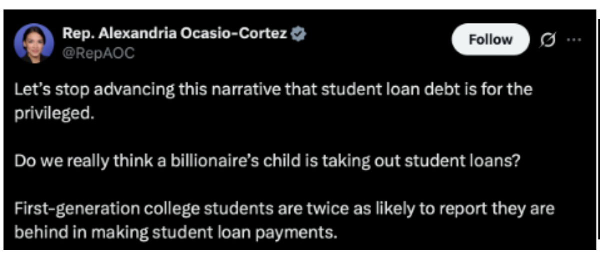

Figure 1: Rep. Alexandria Ocasio-Cortez

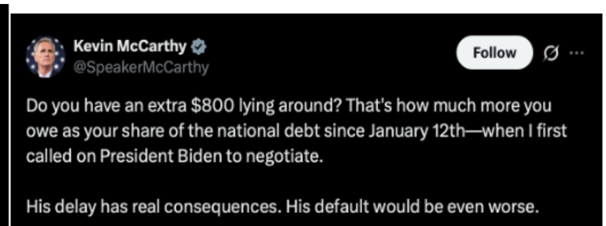

Figure 2: Rep. Kevin McCarthy

3.3 Biases and Limitations

While this classification provides simplicity and replicability, it also introduces potential sources of bias. These fall into two main categories:

(1) Dictionary Limits: Precision vs. Coverage

As noted, the inherent challenge in using a dictionary-based method is balancing precision with coverage. Dictionary methods risk underestimation by missing subtle or implicit deficit references Alternatively, in cases where one uses both large and speculative dictionaries, the classification strategy may overestimate deficit rhetoric. In practice, I adjusted the specificity of both dictionaries iteratively based on the accuracy of their given outputs —which I assessed manually by checking the data. While bias in either direction is statistically possible, I'd argue that my strategy almost certainly underestimates total deficit rhetoric, since there are theoretically infinite ways to reference the federal debt.

(2) Variation in Rhetoric and Style

Bias may also arise from differences in rhetorical style across time, parties, and individuals. For instance, members who consistently use explicit terms like *"national debt"* are captured reliably, while those who reference deficits more vaguely or implicitly are systematically undercounted. If such variation correlates with party or institutional context (e.g., legislative periods), it could bias comparisons.

3.4 Model Improvement: A Machine Learning Approach

For future research considerations—whether involving similar or identical data—it is important to consider alternative, complementary approaches to improve labeling accuracy.

My current strategy relies on a rule-based dictionary method, where I defined Tier 1 and Tier 2 terms and applied them directly to the full dataset. While transparent, and easily adjustable, this approach inevitably misses implicit deficit references and requires subjective judgment in constructing the dictionaries.

A clear improvement would be to use a supervised dictionary validation approach. This method first requires manually labeling enough positive-class instances (i.e., deficit-related tweets), capturing both explicit and implicit references. Because deficit tweets are rare (~1–3% of all tweets), one would need to label roughly 20–30k tweets under an active learning strategy—or 100k+ under random sampling—to obtain 2–3k positive instances. This threshold aligns with best practices for addressing class imbalance (Japkowicz & Stephen 2002; He & Garcia 2009). After constructing this training data, one would build and test multiple dictionary configurations, each with varying levels of specificity, and then select the configuration with the highest validated accuracy.

Accuracy would be assessed with two measures:

(1) Coverage: the share of true deficit tweets correctly labeled.

(2) Precision: the false-positive rate.

Selecting the best model would thus require one to determine the ideal "accuracy" metric, or determining the relative weight of coverage vs precision. This decision—similar to my dictionary choices—requires intuition and could be adjusted based on the researcher's preference.

In sum, a supervised dictionary validation approach would reduce under-detection of implicit references but would not fully eliminate the two forms of bias referenced in 3.2. For instance, expanding the dictionary to improve coverage risks a higher false-positive rate. The training dataset would capture less explicit tweets, and the best-performing dictionary would detect many of these indirect cases. Yet the same expansion may also misclassify unrelated tweets, raising the false-positive ratio. Thus, even if the most comprehensive model achieved the greatest overall accuracy, the fundamental bias would persist, albeit at a smaller scale.

3.5 Other Data

To contextualize congressional tweeting among broader political and economic phenomena, I integrated a wide set of temporal covariates into the dataset. Economic indicators—sourced from the U.S. Treasury and Federal Reserve—include ten-year interest rates, inflation (U.S CPI YoY) and multiple measures of federal debt (publicly held debt, intragovernmental holdings, and total debt outstanding).

Institutional context is captured through variables on presidential party, House and Senate control, majority and minority leadership, and whether government is unified or divided, as well as monthly congressional approval ratings.

Legislative periods are represented by data on major bills from 2017 to 2023 (e.g., CARES Act, PPP, CHIPS Act, IRA). Periods are defined by one month before and after passage. For instance, the Tax Cuts and Jobs Act was signed into law by President Trump in December 2017 making its designated legislative period November 2017 - January 2018. Fiscal impact estimates for each bill come from the Congressional Budget Office (CBO) and are normalized when included in regression specifications.

Finally, I construct a partisanship score to capture the level of partisan division in each legislative period. The measure is defined as the absolute difference between Democratic and Republican support rates for a given bill:

$$\text{Partisanship Score} = \left| \frac{\text{Dem Yea}}{\text{Dem Total}} - \frac{\text{Rep Yea}}{\text{Rep Total}} \right|$$

Figure 3: Partisanship score definition.

This score ranges from 0 (perfect bipartisan support) to 1 (completely partisan support, with only one party voting in favor). For regression analysis, I normalize the score to standardize across legislative periods.